



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 10-403-1
19-10-2021

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SPECIFICATION

FOR

PIN-TYPE POLYMER INSULATOR

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1. SCOPE

This specification describes the minimum technical requirements for design, engineering, manufacturing, testing, inspection, packing, and performance of outdoor Pin-type polymer insulators. The required insulators are intended to be installed in the overhead lines of the medium voltage distribution networks (12 kV and 24 kV) of the Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification, the offered polymer insulators should be designed, manufactured and tested according to the latest editions of the relevant applicable standards given in Table (1).

Table (1)

Standard No.	Description
IEC 60575	Thermal-mechanical performance test and mechanical performance test on string insulator units
IEC 60815	Selection and dimensioning of high-voltage insulators intended for use in polluted conditions
IEC 60383	Insulators for overhead lines with a nominal voltage above 1000 V
IEC 61109	Insulators for overhead lines - Composite suspension and tension insulators for A.C. systems with a nominal voltage greater than 1000 V - Definitions, test methods and acceptance criteria
IEC 62217	Polymeric HV insulators for indoor and outdoor use - General definitions, test methods and acceptance criteria
IEC 61952	Insulators for overhead lines - Composite line post insulators for A.C. systems with a nominal voltage greater than 1000 V - Definitions, test methods and acceptance criteria
IEC 60273	Characteristic of indoor and outdoor post insulators for systems with nominal voltages greater than 1000 V
IEC TR 62039	Selection guide for polymeric materials for outdoor use under HV stress
IEC 60437	Radio Interference Test on High-Voltage Insulators



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3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required insulator should be guaranteed under the environmental conditions given in Table (2). Any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Ambient temperature	-5°C to +45°C (50°C as option according to ...EDC) Very intensive sunlight of 70 °C exposure temperature is expected.
Maximum relative humidity	Up to 100 %
Altitude	Up to 1000 m above sea-level
Pollution	Class D (heavy)
Average thunder storm	45 Days per annum.
Climatic condition	Moderately hot and humid tropical climate, conductive to rust and fungus growth.
Maximum annual rainfall	250 mm
Maximum wind velocity	55 km/hour
Solar energy radiation	$\geq 1100 \text{ W/m}^2$

4. SYSTEM CHARACTERISTICS

The required insulator should have a safe electrical insulation behavior for the electrical system given in Table (3).

Table (3)

Nominal voltage	11 and 22 kV
Highest system voltage	12 and 24 kV
Power frequency	50 Hz
Number of phases	Three-phase
System earthing	Solidly earthed



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5. TECHNICAL REQUIREMENTS

5.1 General:

- The pin-type composite silicone rubber insulators shall meet all requirements of this specification in all respects and shall consist of a single piece of silicon rubber together with the core and its metal parts to form a rigid assembly.
- Manufacturer's drawings shall show the outline of the insulators together with all pertinent dimensions. Any variation in these dimensions due to manufacturing tolerances shall be indicated.
- The tolerances on dimensions (e.g. diameter, length and creepage distance) shall be in accordance with IEC 61109. However, negative tolerance shall not be applicable to creepage distance. The tolerance on dimension d (in millimeters) is estimated as:
 $\pm (0.04d + 1.5) \text{ mm}$ when $d \leq 300 \text{ mm}$
 $\pm (0.025d + 6) \text{ mm}$ when $d > 300 \text{ mm}$ with a maximum tolerance of $\pm 50 \text{ mm}$,
- The design shall be such that stresses due to expansion and contraction in any part of the insulator shall not lead to any deterioration. Precautions shall be taken to avoid any chemical reaction between insulator parts and fitting materials
- The insulators shall be suitable for overhead line conductors of cross-sectional area starting from 35 mm^2 up to 300 mm^2 .
- Pin-type insulators shall be designed to meet the quality, safety, reliability and capability to withstand a wide range of environmental conditions:
 - Core: The internal insulating part
 - Housing: The external insulating part.
 - Metal end fittings: For attaching hardware to support conductor.
- The insulator shall be radio interference free at operating voltage.



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- The insulators shall be made of best quality material free from flaws, cracks, voids, conducting impurities and chipped edges.
- The complete insulator with fittings shall not create excessive or localized corona at a voltage less than its rated voltage.
- The design of insulator and its fitting shall minimize local corona formation and discharges causing radio interferences.
- The electrical and mechanical characteristics for pin-type insulators shall be as given in Table (4).

Table (4)

1	Maximum system voltage (kV)	12	24
2	Operating voltage (kV)	11	22
3	Nominal Frequency (Hz)	50	
4	Wet power frequency withstand voltage (kV)	≥ 45	≥ 90
	Wet power frequency flashover voltage (kV)	≥ 50	≥ 100
5	Dry power frequency withstand voltage (kV)	≥ 60	≥ 115
	Dry power frequency flashover voltage (kV)	75	125
6	Impulse 1.2 / 50 μ s withstand voltage (kV)	≥ 95	150
	Impulse 1.2 / 50 μ s flashover voltage (kV)	110	170
7	Leakage distance not less than (mm)	480	960
8	Minimum Cantilever load (kN)	10	

5.2 Profile:

- The pin-type polymer insulator shall have aerodynamic profile sheds with good hydrophobic and recovery properties. Additionally, it should be maintenance-free in accordance to IEC 60815-3.



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- The sheds shall have alternating profile of uniform spacing between sheds. The shed shape, spacing and inclination shall comply with IEC 60815-3 and shall be able to withstand the different types of pollution.

5.3 Construction:

- Pin-type silicone rubber insulator must be manufactured to form one-piece unit. The insulator shall have minimum breaking load 10 kN.

5.4 Core:

- The pin-type composite silicone rubber insulator has a solid Fiber Reinforced Plastic (FRP-rod) core with silicone rubber housing. The diameter of core and housing should not be less than 38 mm.
- The FRP-rod core shall be epoxy resin reinforced with Electrical Corrosion Resistance (ECR), glass fibers and free of boron. It shall be resistant against acid attacks in order to prevent stress corrosion of the FRP-rod core.

5.5 Housing:

- The housing shall be made of composite silicone-elastomer-rubber compound with a thickness greater than 3.5 mm over the sheath. It shall be one-piece using compression or injection molding principle to cover the core.
- The base polymer shall be silicone rubber prior to the addition of reinforcing fillers.
- The housing shall be made of High Temperature Vulcanizing (HTV) silicone rubber material.
- The weather sheds shall be made of silicone elastomer compound molded as part of the sheath and shall be free from imperfections. The weather sheds should be the same silicone of the housing. It should protect the FRP-rod against environments influences, external pollution and humidity.



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- Sheds and sheath shall be produced in one molding step. Shed-sheath interface is unacceptable. The sheath must be homogeneous around the FRP-rod.
- Sheds shall be characterized by an aerodynamic profile of alternating sheds comply with the requirements of IEC 60815-3. The diameter of insulator sheds shall not exceed 160 mm.
- Based on 40 mm/kV, the creepage distance of the polymeric insulator shall be as follows:
 - For 12 kV insulators: the minimum creepage distance is 480 mm.
 - For 24 kV insulators: the minimum creepage distance is 960 mm

5.6 Metal end Fittings:

- End fittings transmit the mechanical load to the core. The top part should be aluminum alloy suitable for different sizes of aluminum conductors and bottom pin should be galvanized steel. The material used in fittings shall be corrosion resistant.
- Hardware of respective specified mechanical load should be hot-dip galvanized and Zinc coated with minimum 99.95 % purity of electrolytic high grade Zinc. The average thickness of galvanization should not be less than 85 μm for pins. It should not be less than 45 μm for nuts and washers.
- Metal end fittings shall have smooth and uniform surfaces without any sharp edges or corners. It shall be free of cracks, flakes, slag, blow-holes shrinkages defects and localized porosity.
- They shall be connected to the rod by means of a controlled compression technique.
- As the main duty of the end fittings is the transfer of mechanical loads to the core, the fittings should be properly attached to the core by a coaxial or hexagonal compression process and should not damage the individual fibers or crack the core.



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- The finished surface shall be smooth with a good performance. The surface shall not crack or get chipped due to aging effect under normal and abnormal service conditions.
- Bottom end metal fitting (Shank) of pin-type insulator should be as per IEC standard. Minimum length of shank should be 175 mm with unscrewed part not more than 50 mm for all ranges. Minimum Shank diameter should be 22 mm for all ranges. Minimum Collar diameter should be 50 mm and its thickness should not be less than 6 mm. Two nuts and 5 mm thick spring washer shall be supplied.
- Fittings should be effectively sealed to prevent moisture ingress. Effectiveness of sealing system must be supported by test documents.
- All surfaces of the metal parts shall be perfectly smooth without projecting points or irregularities, which may cause corona.
- All load bearing surfaces shall be smooth and uniform enough to distribute the loading stresses uniformly.
- Fixing of conductor to the top of pin insulator should be either with suitable cap or by top-ties and to be supplied with the pin-type insulator.
- Pin size should be M22

6. PACKING and MARKING

- The insulators should be packed in suitable wooden crates, cases, or safe carton boxes to withstand shipment, loading, handling and storing.
- Each insulator shall bear a permanent marking in English language in accordance with IEC standard on the end of fitting.
- All the characters shall be legible, durable and permanently marked on one end fitting or on the uppermost shed of the insulator.
- The use of stickers/labels is not permitted.



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- Each insulator shall bear the following information:
 - a) Breaking failing load.
 - b) Manufactures names
 - c) Year of manufacture-
 - d) Order number
 - e) Type and model of insulator
 - f) Specified cantilever load.
 - g) Rated voltage.
- Crate Marking (English and/or Arabic):
 - a) Nominal system voltage
 - b) Type and No. of insulators
 - c) Order number
 - d) Weight, kg.
 - e) Manufactures names and Name of the manufacturing country.

7. TESTING and INSPECTION

- All tests shall be carried out by and at the expense of the supplier in accordance to the latest relevant standard.
- ...EDC reserves the right to visit the factory during manufacture of any or all items covered by this specification, for inspection of material or witness of tests. Accordingly, the manufacturer shall give ...EDC testing schedule.

7.1 Design Tests:

- Polymer pin-type insulator shall be tested according to IEC 62217 for the design tests given in Table (5).



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Table (5)

No.	Test Description	IEC	Clause No.
1	Test on interfaces and connection of end fittings		
1.1	Thermal Mechanical pre-stress	61952	10.3.1
1.2	Water immersion test	62217	9.2.6
1.3	Verification test	62217	9.2.7
1.4	Visual examination	62217	9.2.7.2
1.5	Steep front impulse voltage	62217	9.2.7.3
1.6	Dry power frequency voltage test	62217	9.2.7.4
2	Test on shed and housing material		
2.1	Hardness test	62217	9.3.1
2.2	Accelerated weathering test	62217	9.3.2
2.3	Tracking and erosion test	62217	9.3.3
2.4	Flammability test	62217	9.3.4
3	Test on core material		
3.1	Dye penetration test	62217	9.4.1
3.2	Water diffusion test	62217	9.4.2
4	Assembled core load time test		
4.1	Tensile load test	61952	10.4.2
4.2	Verification of the maximum design cantilever load (MDCL)	61952	10.4.2.1
5	Hydrophobicity	62073	6
6	Acid resistance of rod	62039	3.8

7.2 Type Tests:

- Polymer pin-type insulator shall be tested for the following electrical tests:
 - a) Dry lightning impulse withstand voltage test (IEC 61952 cl. 11.1)
 - b) Wet power-frequency test (IEC 61952 cl. 11.1)
 - c) Cantilever failing load test (IEC 61952 cl. 11.2)



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7.3 Sample Tests:

- Polymer pin-type insulator shall be tested as per Clause 12 of IEC 61952 for the following sample tests:
 - a) Verification of dimensions. (E1+E2)
 - b) Verification of the Specified Cantilever Load (SCL). (E1)
 - c) Galvanizing Test. (E1+E2)

Lot size (N)	Sample size	
	E1	E2
$N \leq 100$	By agreement	
$100 < N \leq 300$	2	1
$300 < N \leq 2\,000$	4	3
$2\,000 < N \leq 5\,000$	8	4
$5\,000 < N \leq 10\,000$	12	6

7.4 Routine Tests:

- Polymer Pin-type insulator shall be tested as per Clause 13 of IEC 61952 for the following routine tests:
 - a) Tensile load test.
 - b) Visual examination.

8. GUARANTEE

- The supplier guarantee the supplied materials against all defects arising out of faulty design or workmanship, or of defective material for a period of five years at least from date of delivery.
- Any defective parts of poor-quality or poor-assembly materials that are detected during the guarantee period should be replaced or repaired as soon as possible and free of charge.



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9. SUBMITTALS

9.1 Submittals required with tender:

- The supplier shall complete and return one copy of the attached Data Schedule.
- Technical data and original catalogues shall be submitted with the offer.
- One specimen of the offered insulator shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the material they intend to supply. Dimensional verification, indelible markings and finishing shall be checked.
- All necessary drawings including layout, dimensions, should be submitted with the offer. The drawing should include but not be limited to the following information:
Nuts and spring washer shall be hot dip galvanized.
- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Copy of type test reports.
- List of deviations & clauses to which exception is taken (if any).

9.2 Submittals required following award of contract:

- Details of manufacturing and testing schedules
- Routine test reports
- The supplier should provide all the instructions of use/storing in (Arabic and English).

10. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.



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Technical Data Schedule

NO	ITEMS	Vendor specifications
1	Nominal system voltage	kV
2	Highest system voltage	kV
3	Frequency	 Hz
4	Lightning impulse withstand voltage 1.2/50 μ s (LIWL) kV peak	 kV
5	Minimum specific leakage path based on highest voltage	 mm/kV
6	Specified cantilever load	kN
7	Dry power frequency flashover voltage	 kV
8	Wet power frequency flashover voltage	 kV
9	Dry power frequency withstand voltage	 kV
10	Wet power frequency withstand voltage	 kV
11	Diameter of shed (large / small)	mm /mm
12	Diameter of core with housing	mm
13	Minimum thickness of housing	 mm
14	Withstand bird biting	Yes/No
15	Color of insulation	
16	Total length	mm
17	Weight of insulator	kg
18	Marking as per clause 6	Yes/No
19	Type of material components: - Rod - Weather sheds - Bottom Fitting - Top Fitting	
20	Guarantee period	year

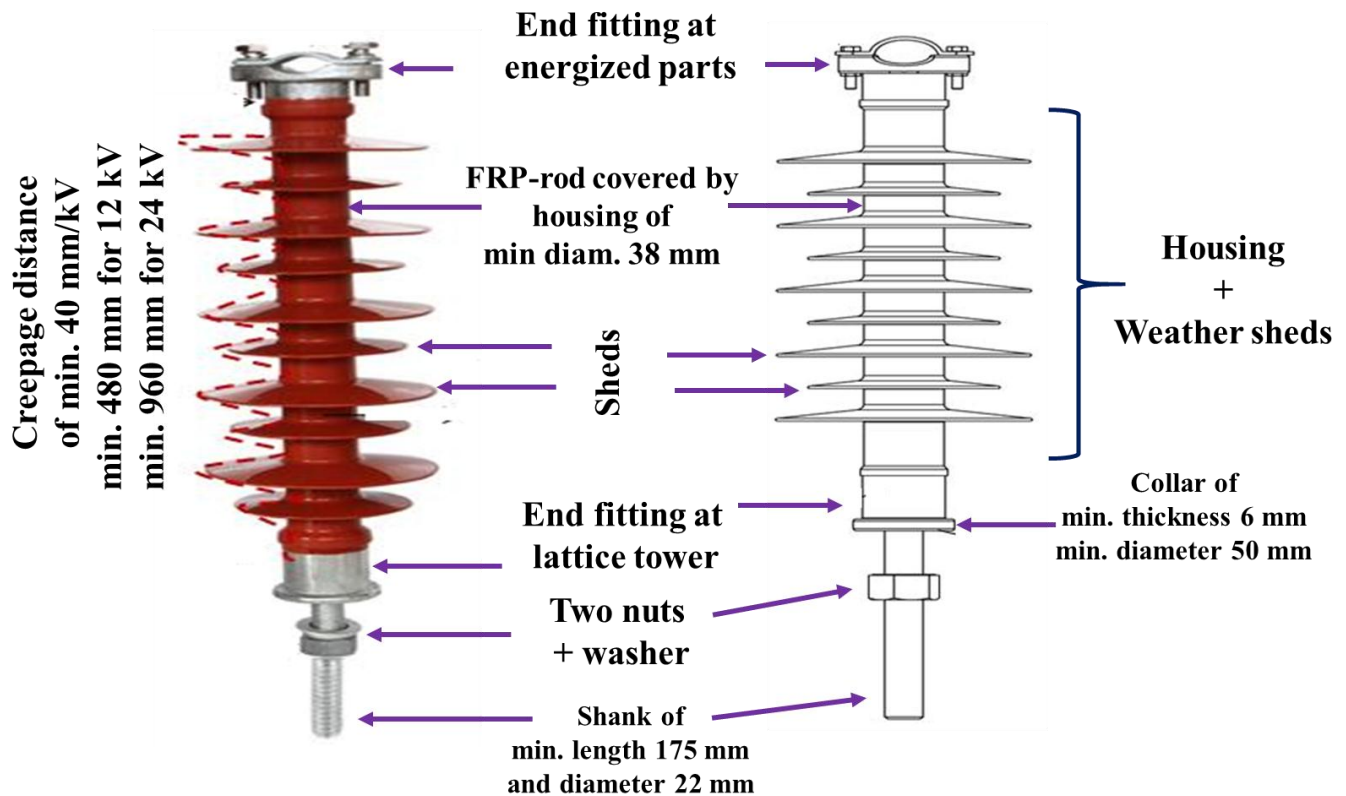
We guarantee the data given above for the offered insulator toEDC.

Signature: Date:

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Pin-type Polymer Insulator 12 and 24 kV



- c is the minimum distance between adjacent sheds of the same diameter.
- Negative tolerance shall not be applicable to creepage distance.
- The tolerance on dimension d (in millimeters) is:
 $\pm (0.04d + 1.5)$ mm when $d \leq 300$ mm
 $\pm (0.025d + 6)$ mm when $d > 300$ mm with a maximum tolerance of ± 50 mm,

